6.4.3 Application of Transformer

STE9CS

- It is used to step up or step down the voltages and currents in an electrical circuit.
- Used in modern electrical transmission and distribution systems.
- It boost voltage levels so as to decrease line losses during transmission.

6.5 TYPES OF INSTRUMENT TRANSFORMERS

6.5.1 Potential Transformer

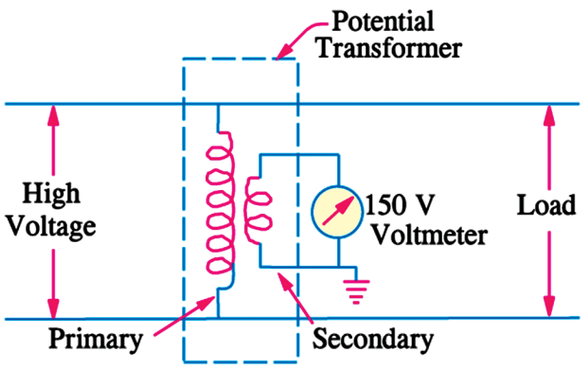


Fig 6.8 *Potential Transformers*

Potential transformer do not differ much from the ordinary two windings transformers (Refer Figure 6.8). These are step down transformers. The primary winding is connected directly across the power circuit. The secondary is usually rated for 110 (or) 220 volts. Voltage ratio is depenting upon primary voltage .

6.5.2 Current Transformer

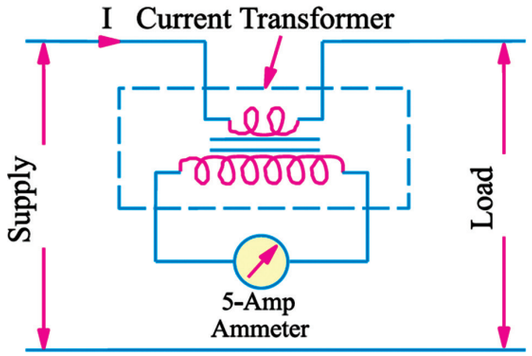


Fig 6.9 *Current Transformers*

The current transformer has a primary coil of only few turns of thick wire connected in series with the line whose current is to be measured. The secondary coil consisting large number of turns is connected to the terminals of a low range ammeter (See Figure 6.9). The ammeter is thus entirely insulated from the line. Mostly the secondaries of all C.T are wound for 5 amperes.

6.5.3 Auto Transformer (VARIAC)

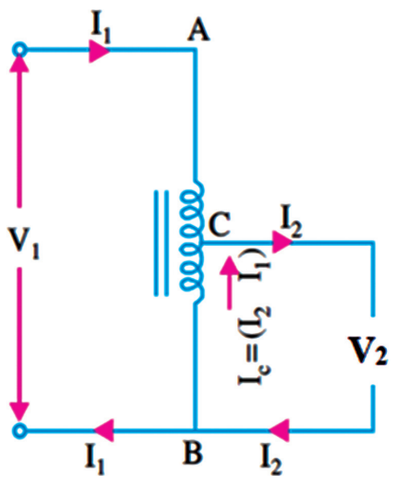


Fig 6.10 *Auto Transformer*

An auto transformer is a single winding transformer which is used to get varying AC voltage. Consider a single winding BA of N_1 turns wound on an iron core as shown in Figure 6.10. If core loss, copper loss, magnetizing current and leakage reactance are neglected. If this winding

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